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Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Aim

The aim of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Dataset Description

Dataset: TensorFlow Flowers (`tf_flowers`)

The dataset contains 3670 natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images vary in scale, viewpoint, illumination, background clutter, flower size, color, texture and object location, which makes the dataset suitable for studying attention.

2.1 Image Preprocessing

Every input image is resized to $224 \times 224 \times 3$ and preprocessed with the procedure expected by the ImageNet-pretrained ResNet50. Data augmentation: random horizontal flipping, small rotations, random zoom, random translation and mild contrast variation. The same augmentation is maintained for all model comparisons.

2.2 Dataset Split

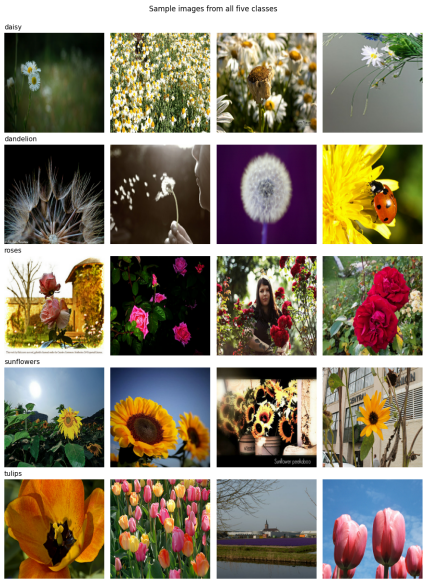
The same stratified split is used for every experiment:

$\boxed{70\% : 15\% : 15\%}$ for Training : Validation : Testing.

Class	Train	Validation	Test	Total
daisy	443	95	95	633
dandelion	629	134	135	898
roses	449	96	96	641
sunflowers	489	105	105	699
tulips	559	120	120	799
Total	2569	550	551	3670

The exact same samples are retained in each subset for all ablation experiments, so differences in performance are caused by the model and not by the data split.

Sample Images from TensorFlow Flowers



Inference: The classes differ in colour and flower size and many images contain heavy background clutter which is where attention could help or mislead. The class proportions are similar in all three splits.

3. ResNet50 Architecture

ResNet50 pretrained on ImageNet is the **only backbone**. For an input $I \in \mathbb{R}^{224\times224\times3}$ the major stages produce:

ResNet50 Stage	Spatial Resolution	Channels
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper, spatial resolution decreases and feature channels increase, moving from low-level to semantic information. Residual connections ($y = F(x) + x$) ease training of the 50 layers.

- **Early features:** edges, colors, orientations, boundaries and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements and shape.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may behave differently at different locations.

4. Attention Mechanisms

Let the input feature tensor be $F \in \mathbb{R}^{H\times W\times C}$.

4.1 Channel Attention

Global average pooling gives $z_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F(i,j,c)$, the channel weights are

$$M_c(F) = \sigma\left(W_2\,\delta(W_1 z)\right), \qquad F_c = F \odot M_c(F),$$

with reduction ratio $r = 16$. It answers: **which feature channels are important?**

4.2 Spatial Attention

With $F_{\text{avg}} = \text{Mean}_c(F)$ and $F_{\max} = \text{Max}_c(F)$,

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}\left([F_{\text{avg}} \| F_{\max}]\right)\right), \qquad F'_s = F \odot M_s(F).$$

It answers: **which spatial locations are important?**

4.3 Sequential Channel–Spatial Attention

$$F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}$$

first *what*, then *where*. The reverse order is Spatial→Channel.

4.4 Lightweight Self-Attention

With $X \in \mathbb{R}^{N\times C}$, $N = HW$, $Q = XW_Q$, $K = XW_K$, $V = XW_V$:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \qquad Y = AV, \qquad F_{\text{SA}} = F + \gamma\,\text{reshape}(Y).$$

The $N \times N$ matrix makes it expensive at high resolution, so it is used at P_3.

4.5 Additional Mechanisms

SE (same as channel attention A1), **ECA** (1-D convolution over the channel descriptor, no dimensionality reduction), **CBAM** (channel attention with average and max pooling, then a 7×7 spatial gate) and **Coordinate attention** (direction-aware pooling along H and W).

Module at P_3	Channel/SE	Spatial	Ch+Sp	Self	Combined	ECA	CBAM	Coord.
Attention parameters	132,160	99	132,259	1,312,001	1,444,260	5	132,259	100,512

5. Proposed Experimental Architecture

The baseline model is:



For an attention model the block is inserted at a chosen stage:

Previous ResNet Stage → Attention → Next ResNet Stage

e.g. attention at P_3: conv4_x → Attention → conv5_x. The recorded feature dimensions before and after the attention block (identical in all 19 inserted blocks) are:

Position	**Location**	**Resolution**	**Channels**	**Dims before = after**
P_1	After conv2_x	56×56	256	(56,56,256)
P_2	After conv3_x	28×28	512	(28,28,512)
P_3	After conv4_x	14×14	1024	(14,14,1024)
P_4	After conv5_x	7×7	2048	(7,7,2048)

6. Experimental Setup

All controlled comparisons use the same dataset split, preprocessing, augmentation, backbone, classifier, optimizer, batch size, number of epochs, learning-rate schedule, fine-tuning procedure and evaluation metrics. Only the experimental factor under study is changed.

6.1 Transfer-Learning Strategy

1. Load ResNet50 with ImageNet weights and remove the original classifier.
2. Insert the selected attention module, then Global Average Pooling, Dropout and a Dense five-class layer.
3. Freeze the pretrained layers and train the new layers.
4. Unfreeze the final ResNet50 stage (conv5_x) and fine-tune with a smaller learning rate.

Setting	Value Used
Input size	$224 \times 224 \times 3$
Backbone	ResNet50, ImageNet pretrained
Dataset split	70% / 15% / 15% (stratified)
Optimizer	Adam
Initial learning rate	10^{-3}
Fine-tuning learning rate	10^{-4} (BatchNorm frozen)
Batch size	32
Epochs	15 (8 frozen + 7 fine-tuning)
Dropout	0.3
Reduction ratio / spatial kernel	$r = 16$ / $k = 7$
Random seed	42

6.2 Loss Function

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c}.$$

6.3 Evaluation Metrics

Accuracy, macro precision, macro recall, macro F1, trainable and total parameters, and average inference time per image (GPU, batch of 32). For class c:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \quad \text{Macro F1} = \frac{1}{5}\sum_{c=1}^{5} F1_c.$$

7. Attention-Type Ablation

The attention position is fixed at

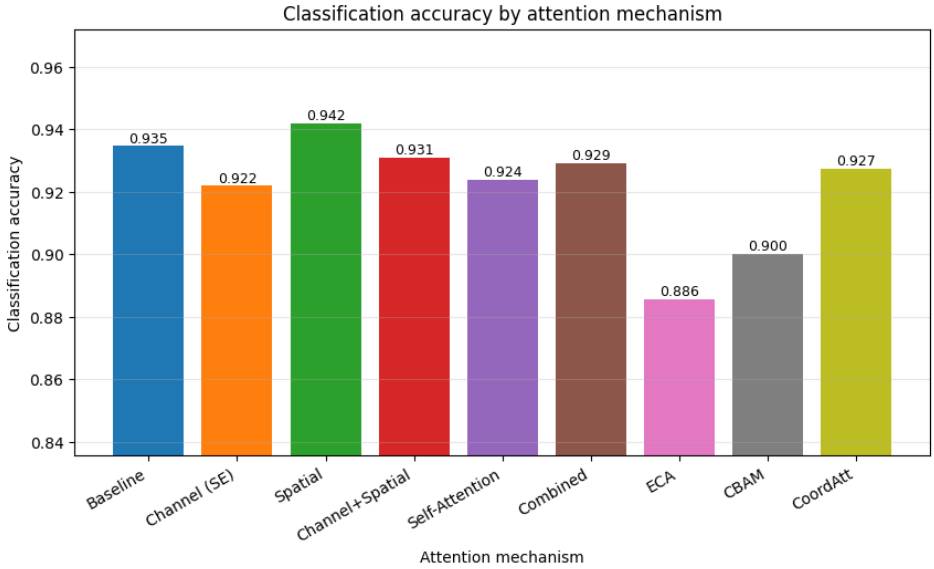
$$\boxed{P_3 = \text{after conv4_x}}.$$

Only the attention mechanism is changed.

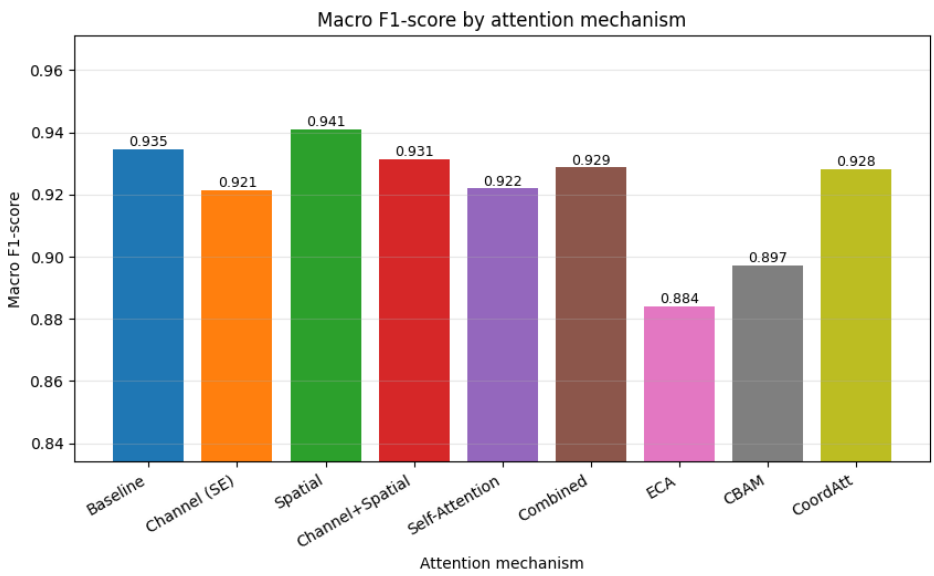
Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

Model	Accuracy	Precision	Recall	Macro F1	Inference Time
Baseline	0.9347	0.9355	0.9341	0.9345	7.25 ms
Channel	0.9220	0.9292	0.9189	0.9213	8.01 ms
Spatial	0.9419	0.9393	0.9449	0.9411	7.44 ms
Channel+Spatial	0.9310	0.9331	0.9312	0.9315	6.95 ms
Self-Attention	0.9238	0.9230	0.9241	0.9219	6.74 ms
Combined	0.9292	0.9325	0.9261	0.9287	7.67 ms

Attention Mechanism vs Test Accuracy



Inference: Spatial attention gave the highest accuracy. Channel, channel+spatial, self-attention and combined attention were slightly below the baseline. ECA and CBAM were lowest .



Inference: Macro F1 follows same like accuracy and stays within 0.004 of accuracy for every model, so errors are spread evenly over the five classes.

8. Attention-Position Ablation

The attention type is fixed as sequential channel–spatial attention and the same module is inserted at different ResNet stages.

Experiment	Position	Resolution	Channels	Test Accuracy	Macro F1
B1	After conv2_x (P_1)	56×56	256	0.9093	0.9089
B2	After conv3_x (P_2)	28×28	512	0.9038	0.9037
B3	After conv4_x (P_3)	14×14	1024	0.9310	0.9315
B4	After conv5_x (P_4)	7×7	2048	0.9347	0.9339

Additional Controlled Ablations

Attention Placement	Accuracy	Macro F1	Attention Params
P_3 only	0.9310	0.9315	132,259
P_4 only	0.9347	0.9339	526,563
$P_3 + P_4$	0.9383	0.9382	658,822
$P_2 + P_3 + P_4$	0.9292	0.9288	692,233

Order / Kernel at P_3	Accuracy	Macro F1	Inference Time
Channel $\rightarrow$ Spatial (7×7)	0.9310	0.9315	6.95 ms
Spatial $\rightarrow$ Channel (7×7)	0.9328	0.9331	7.51 ms
Spatial kernel 3×3	0.9365	0.9365	7.24 ms
Spatial kernel 5×5	0.9274	0.9278	7.28 ms
Spatial kernel 7×7	0.9310	0.9315	6.95 ms

Inference: Accuracy rises with depth (P_1 0.9093, P_2 0.9038, P_3 0.9310, P_4 0.9347); attention on high-resolution/low-level (P_1) and intermediate (P_2) features lowered accuracy by about 2.5–3.1 relative to the baseline (0.9347) while the late semantic positions (P_3, P_4) were comparable to it. $P_3 + P_4$ was best (0.9383) but only +0.36 over P_4 for 5 times the attention parameters and adding P_2 lowered accuracy (0.9292) so more locations are not necessarily better. Channel→Spatial and Spatial→Channel differ by only 0.18 pp, and kernel size gave no useful gain (3×3 best at 0.9365)

9. Performance Results

The main experimental results are summarized below.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	0.9347	0.9355	0.9341	0.9345	23,597,957	7.25 ms
Channel Attention	P_3	0.9220	0.9292	0.9189	0.9213	23,730,117	8.01 ms
Spatial Attention	P_3	0.9419	0.9393	0.9449	0.9411	23,598,056	7.44 ms
Channel+Spatial	P_2	0.9038	0.9221	0.8956	0.9037	23,631,368	7.31 ms
Channel+Spatial	P_3	0.9310	0.9331	0.9312	0.9315	23,730,216	6.95 ms
Channel+Spatial	P_4	0.9347	0.9363	0.9326	0.9339	24,124,520	7.63 ms
Self-Attention	P_3	0.9238	0.9230	0.9241	0.9219	24,909,958	6.74 ms
Additional Attention-1 (ECA)	P_3	0.8857	0.8996	0.8861	0.8841	23,597,962	7.32 ms
Additional Attention-2 (CBAM)	P_3	0.9002	0.9107	0.8949	0.8970	23,730,216	7.93 ms
Additional Attention-3 (Coordinate)	P_3	0.9274	0.9296	0.9270	0.9282	23,698,469	7.65 ms

Overall observation: The baseline reached 93.47% accuracy . Spatial attention was best and Channel+Spatial matched the baseline and Channel+Spatial , ECA and CBAM were clearly lower.

Model	Train acc	Val acc	Train loss	Val loss	Test loss
Baseline	0.9829	0.9327	0.0545	0.3144	0.2040
Channel (SE)	0.9805	0.9218	0.0538	0.3263	0.2517
Spatial	0.9829	0.9309	0.0580	0.3175	0.2386
Channel+Spatial	0.9739	0.9273	0.0682	0.2972	0.2403
Self-Attention	0.9817	0.9255	0.0583	0.3055	0.3031
Combined (Ch+Sp+Self)	0.9739	0.9309	0.1007	0.2584	0.2073
Ch+Sp at P2	0.9790	0.9073	0.0572	0.4089	0.3528
Ch+Sp at P4	0.9887	0.9291	0.0340	0.2779	0.2301
Ch+Sp at P1	0.9782	0.9091	0.0644	0.3557	0.2924
Ch+Sp at P3+P4	0.9848	0.9309	0.0474	0.2755	0.2112
Ch+Sp at P2+P3+P4	0.9763	0.9345	0.0781	0.2919	0.2268
Spatial→Channel	0.9685	0.9345	0.0990	0.2784	0.1988
Ch+Sp, $k = 3$	0.9708	0.9382	0.0883	0.2935	0.1929
Ch+Sp, $k = 5$	0.9798	0.9236	0.0697	0.4145	0.3937
ECA	0.9720	0.8855	0.0860	0.5616	0.6102
CBAM	0.9774	0.9145	0.0585	0.3548	0.3021
Coordinate Attention	0.9638	0.9291	0.0954	0.4183	0.3581

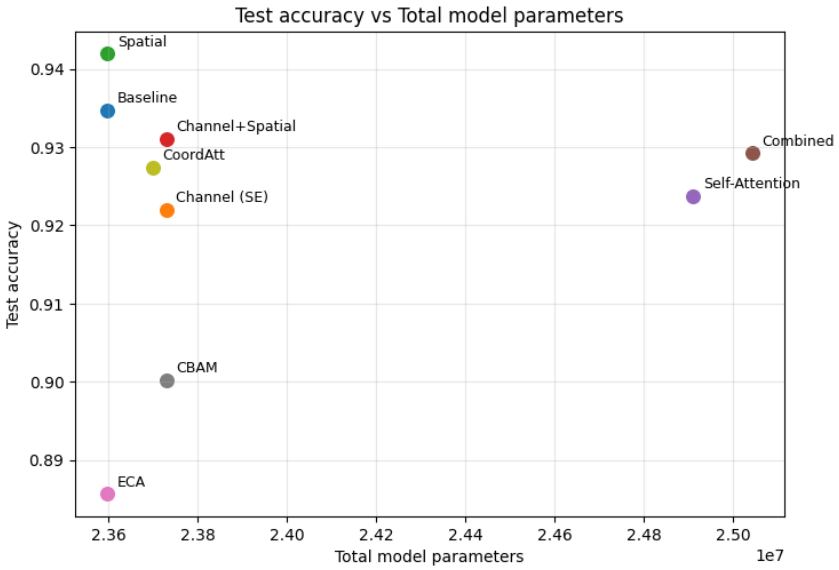
Training accuracy (96–99%) exceeds validation/test accuracy (88–94%) for every model, so all models overfit slightly.

10. Complexity Analysis

Attention should be evaluated not only by accuracy but also by the extra parameters and inference cost it introduces.

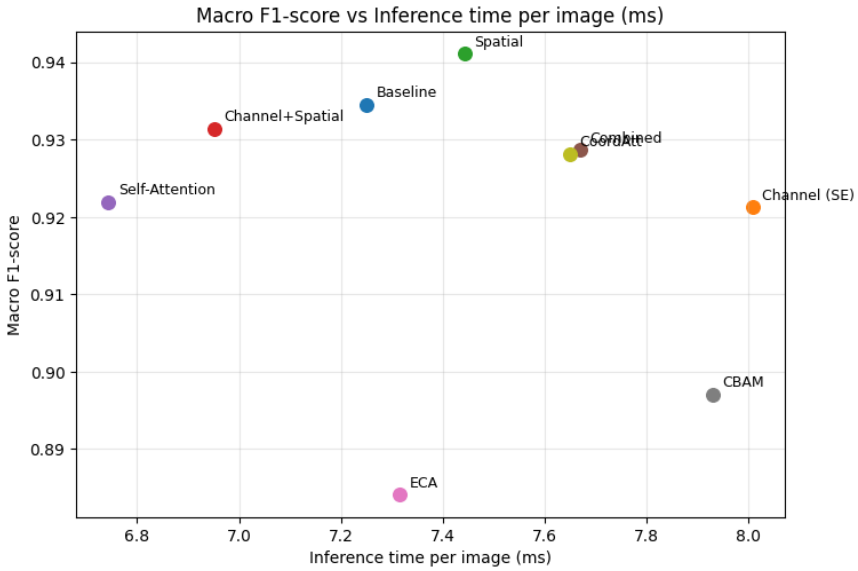
Model	Trainable (Fine-Tune)	Total Parameters	Attention Params	Inference Time / Image	Training Time
Baseline	14,963,717	23,597,957	0	7.25 ms	252 s
Channel (SE)	15,095,877	23,730,117	132,160	8.01 ms	268 s
Spatial	14,963,816	23,598,056	99	7.44 ms	267 s
Channel+Spatial	15,095,976	23,730,216	132,259	6.95 ms	273 s
Self-Attention	16,275,718	24,909,958	1,312,001	6.74 ms	291 s
Combined (Ch+Sp+Self)	16,407,977	25,042,217	1,444,260	7.67 ms	332 s
Ch+Sp at P2	14,997,128	23,631,368	33,411	7.31 ms	313 s
Ch+Sp at P4	15,490,280	24,124,520	526,563	7.63 ms	258 s
Ch+Sp at P1	14,972,280	23,606,520	8,563	7.59 ms	353 s
Ch+Sp at P3+P4	15,622,539	24,256,779	658,822	7.01 ms	279 s
Ch+Sp at P2+P3+P4	15,655,950	24,290,190	692,233	7.88 ms	330 s
Spatial→Channel	15,095,976	23,730,216	132,259	7.51 ms	270 s
Ch+Sp, $k = 3$	15,095,896	23,730,136	132,179	7.24 ms	272 s
Ch+Sp, $k = 5$	15,095,928	23,730,168	132,211	7.28 ms	273 s
ECA	14,963,722	23,597,962	5	7.32 ms	266 s
CBAM	15,095,976	23,730,216	132,259	7.93 ms	272 s
Coordinate Attention	15,064,165	23,698,469	100,512	7.65 ms	283 s

Accuracy vs Model Parameters



Inference: Total parameters span only 23.60M to 25.04M and show no positive relation with accuracy: the best point is Spatial attention (23.60M, 0.9419), while Self-Attention (24.91M) and Combined (25.04M) added 1.3–1.4M parameters and yet reached only 0.9238 and 0.9292.

Macro F1-score vs Inference Time

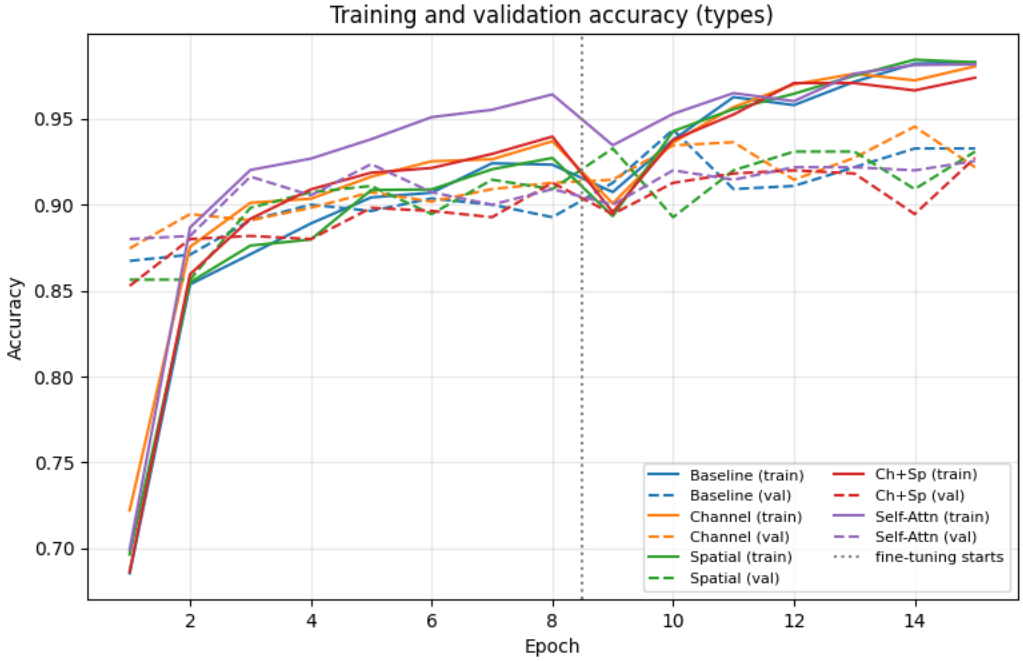


Inference: Inference times lie in a narrow 6.7–8.0 ms per image band with no trend (Self-Attention being the heaviest was measured fastest), so the differences are mostly timing noise .

11. Training and Validation Graphs

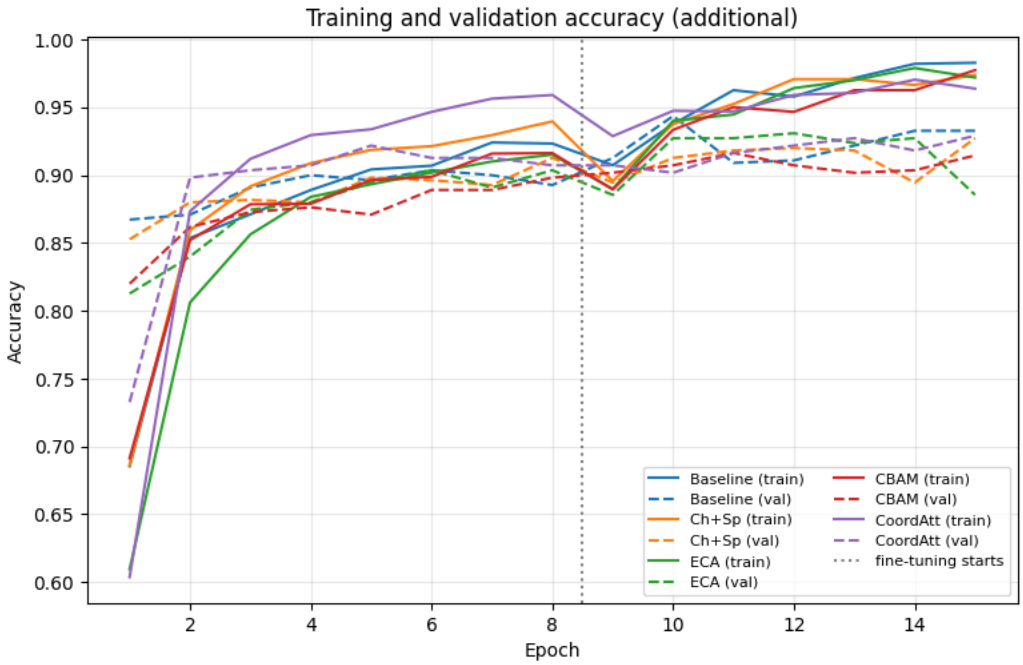
Training and validation curves are used to study convergence, stability and overfitting. The dotted vertical line marks the start of fine-tuning (epoch 9).

Training and Validation Accuracy (Attention Types)



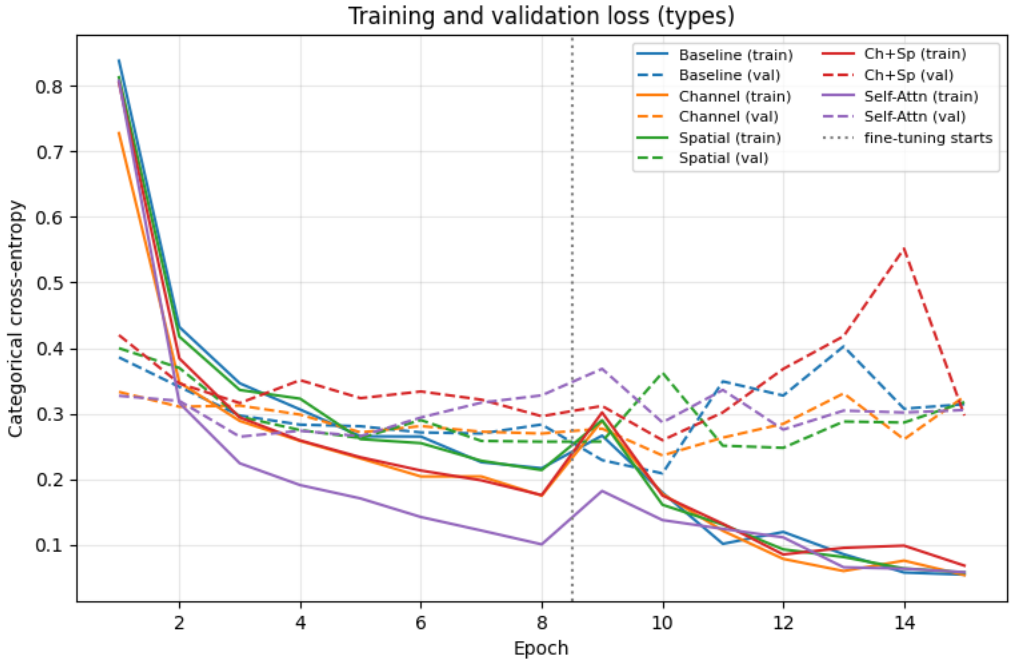
Inference: All models rise from about 0.69 to 0.86 in two epochs and dip when conv5_x is unfrozen; Self-Attention fits the training data fastest (0.96 vs 0.92 at epoch 8) but its validation accuracy ends at the same 0.92–0.93, and the final train–validation gap is 4–6 for all models.

Training and Validation Accuracy (Additional Mechanisms)



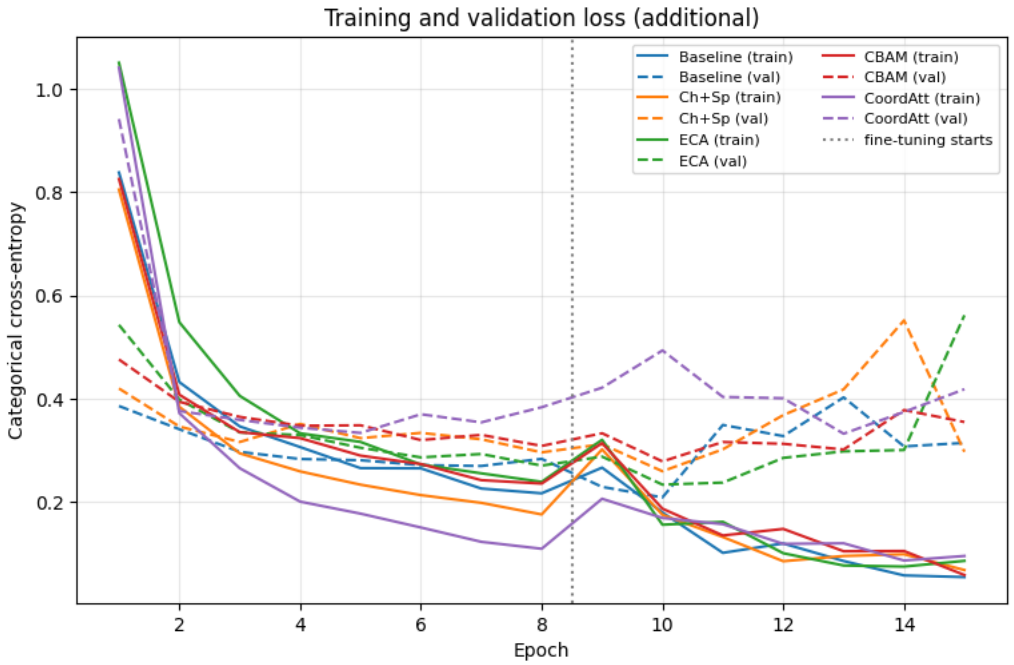
Inference: Coordinate attention fits the training set fastest while ECA starts slowest and its validation accuracy falls to 0.886 in the last epoch; CBAM stays slightly below the baseline in validation accuracy.

Training and Validation Loss (Attention Types)



Inference: Training loss falls to 0.05-0.10 for all models, but validation loss stays at 0.27-0.40 and rises after fine-tuning; Channel+Spatial surges to 0.55 at epoch 14 and Self-Attention has the steadiest validation loss. So attention did not remove overfitting.

Training and Validation Loss (Additional Mechanisms)

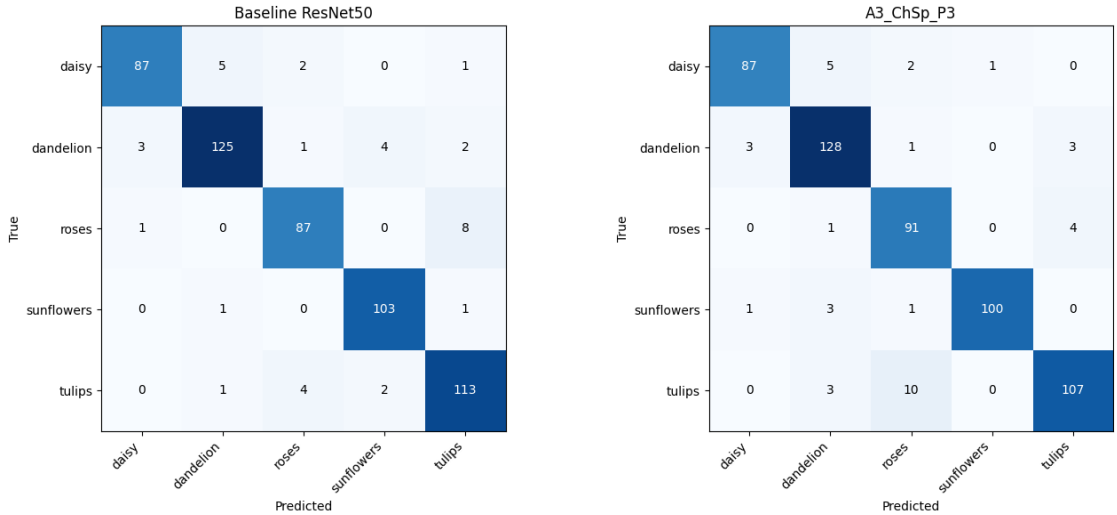


Inference: ECA ends with the highest validation loss (0.56) and Coordinate attention keeps a higher validation loss (about 0.4-0.5) than the baseline despite its low early training loss so it overfits fastest. CBAM stays close to the baseline.

12. Confusion Matrix Analysis

Confusion matrices are generated for the baseline ResNet50 and the selected attention model A3 (Channel+Spatial at P_3).

Baseline and Attention-Model Confusion Matrices



Inference: The baseline correctly classifies 87/95 daisies, 125/135 dandelions, 87/96 roses, 103/105 sunflowers and 113/120 tulips; A3 classifies 87, 128, 91, 100 and 107. Attention improves dandelion (+3) and roses (+4) but reduces sunflower (-3) and tulip (-6) counts; roses predicted as tulips fall from 8 to 4, but tulips predicted as roses rise from 4 to 10, so the gain is class dependent and overall accuracy is lower.

Class	Recall baseline	Recall A3	F1 baseline	F1 A3
daisy	0.9158	0.9158	0.9355	0.9355
dandelion	0.9259	0.9481	0.9363	0.9309
roses	0.9062	0.9479	0.9158	0.9055
sunflowers	0.9810	0.9524	0.9626	0.9709
tulips	0.9417	0.8917	0.9224	0.9145

Sunflowers are classified best; the most frequent confusions are between roses and tulips and daisy and dandelion; sunflowers and roses recall benefit most. A confusion matrix shows which classes are confused which overall accuracy alone cannot.

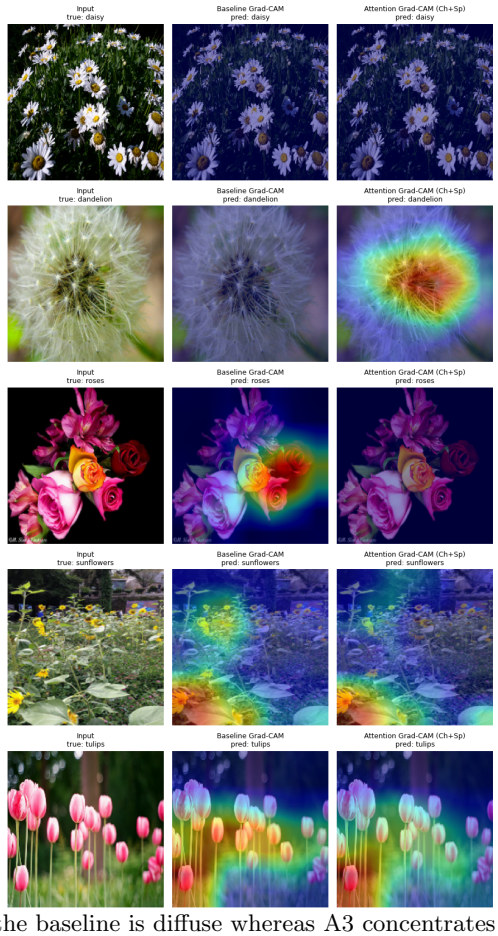
13. Grad-CAM Analysis

Grad-CAM is a post-hoc, class-specific explanation. For feature map A^k and class score y^c,

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

It is computed on the conv5_x output for the predicted class, resized and overlaid on the input; the same test images are used for all models.

Input vs Baseline Grad-CAM vs Attention-Model Grad-CAM

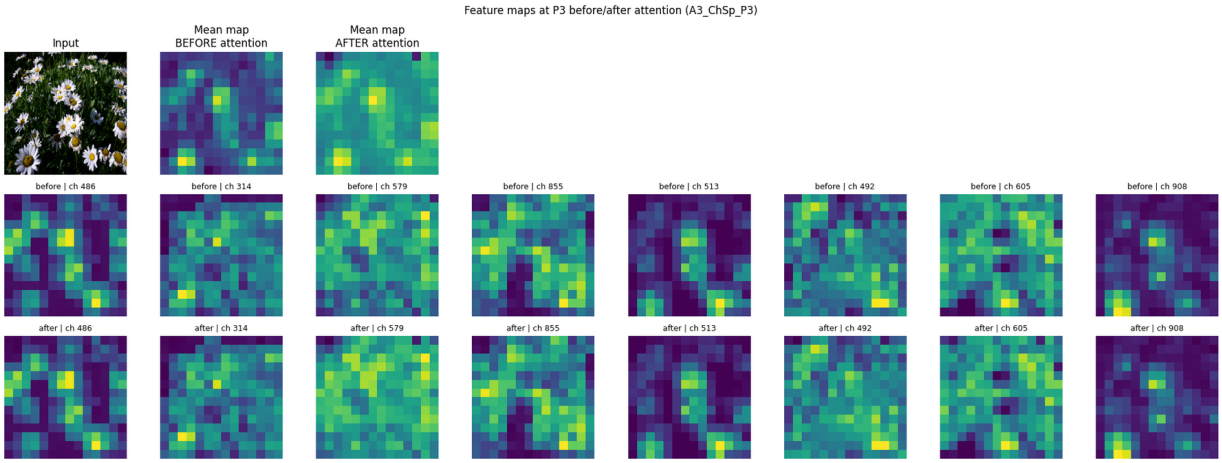


Inference: On the dandelion the baseline is diffuse whereas A3 concentrates strongly on the central head. On tulips and sunflowers both models show similar overall activation (including stems, leaves and ground), and for the rose image the baseline covers the flower cluster while the A3 map is almost empty. So the effect of attention on localization is image dependent, not consistent background suppression .

14. Internal Attention-Map Analysis

Internal attention maps are produced by the attention module during the forward pass, whereas Grad-CAM is generated after prediction using gradients; the two are not the same quantity.

Feature Maps Before and After Attention at P_3

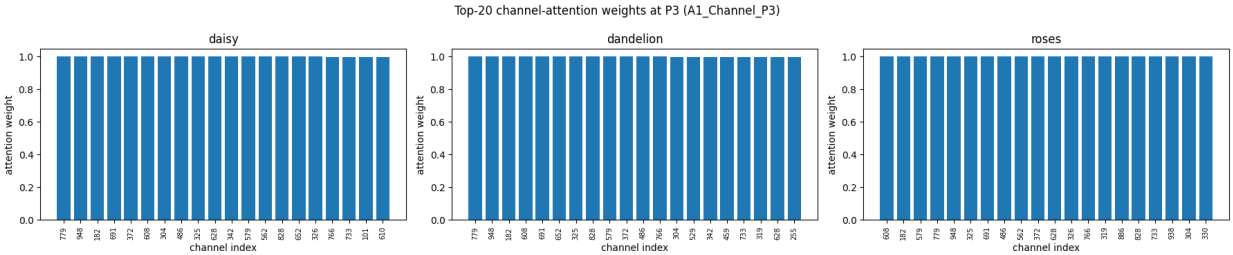


Inference: The activation layout before and after attention is almost identical, and the mean map after attention is only slightly brighter and more uniform, so the gates at P_3 re-weight the features very little .

14.1 Channel-Attention Weights

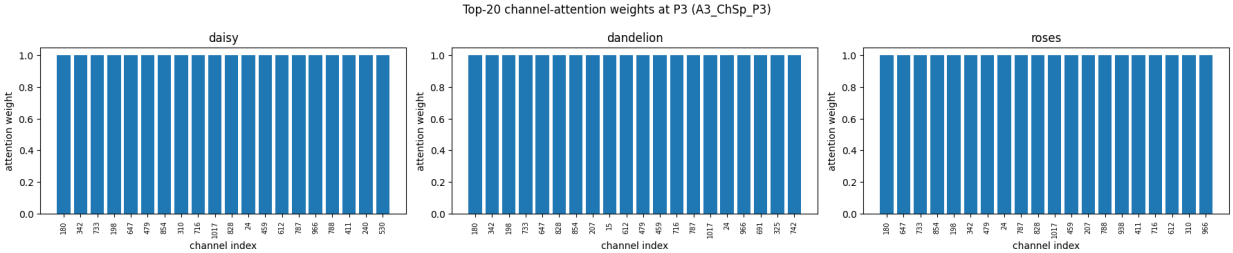
For channel attention $M_c(F) = [m_1, \ldots, m_C]$; the top-$K$ weights ($K = 20$) are shown.

Channel-Attention Weights (A1, Channel at P_3)



Inference: The 20 strongest weights are all saturated at about 1.0 and the same channels (e.g. 779, 948, 182) dominate for the daisy, dandelion and rose images so channel weighting barely changes between images.

Channel-Attention Weights (A3, Channel+Spatial at P_3)

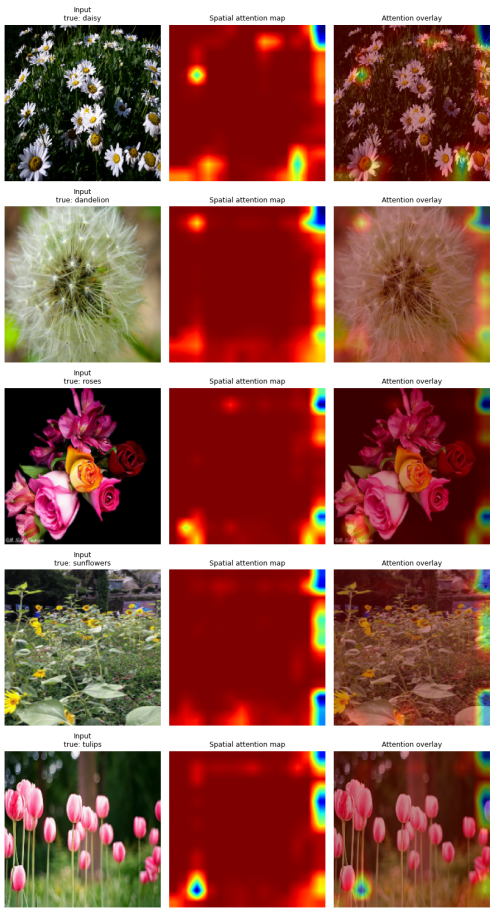


Inference: The top-20 weights are all near 1.0 and the same channels (e.g. 180, 342, 733) dominate for all three classes .

14.2 Spatial-Attention Map and Overlay

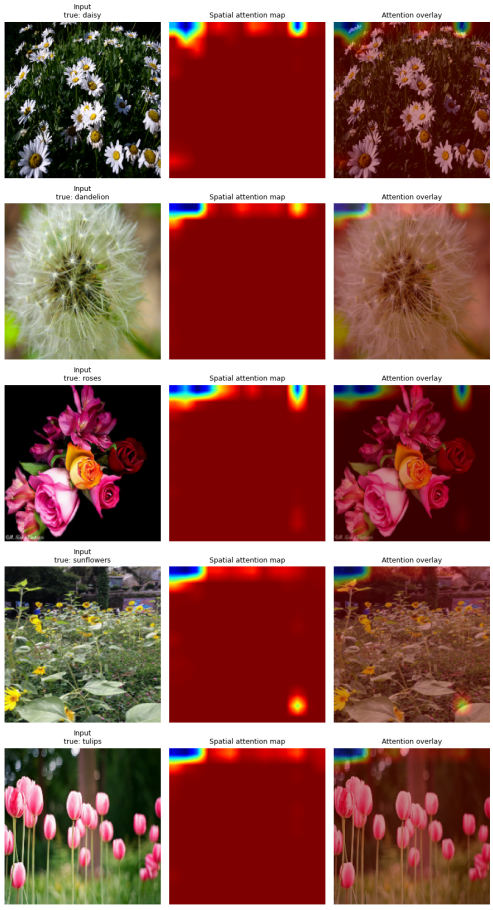
$M_s(F) \in \mathbb{R}^{H \times W}$ is resized to 224×224 and overlaid on the input.

Input vs Spatial Attention Map vs Overlay (A2, Spatial at P_3)



Inference: The map is almost uniformly high and drops only at the borders and in a few small spots on some flowers. It does not separate the flower from the background.

Input vs Spatial Attention Map vs Overlay (A3, Channel+Spatial at P_3)

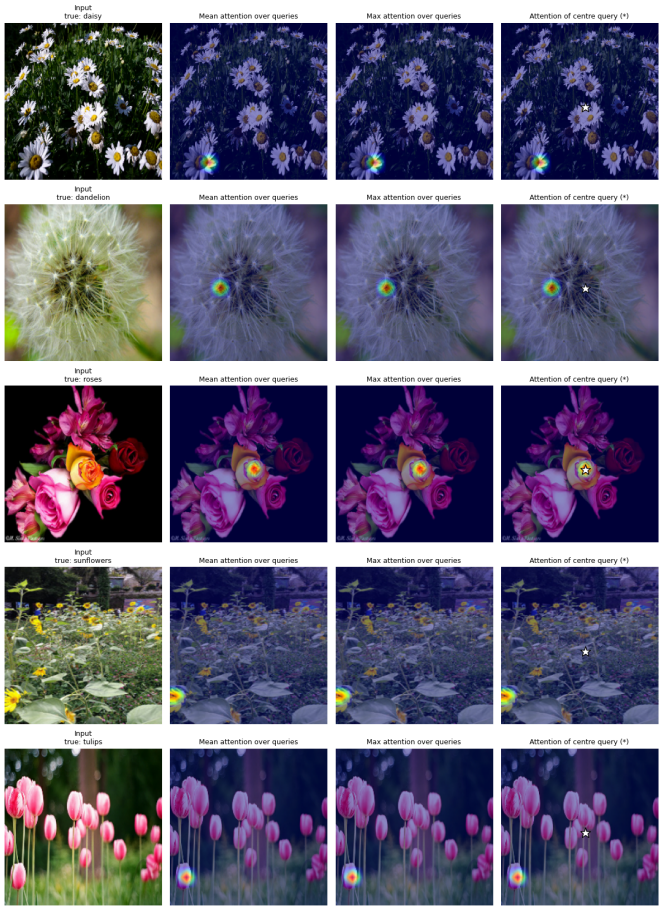


Inference: Attention is high almost everywhere and low only near the image corners/borders, so the learned spatial gate mainly suppresses border locations and is not object oriented.

14.3 Self-Attention Response

Mean attention over queries, maximum attention over queries and the attention of the centre query (marked by a star) are shown.

Self-Attention Response (A4)

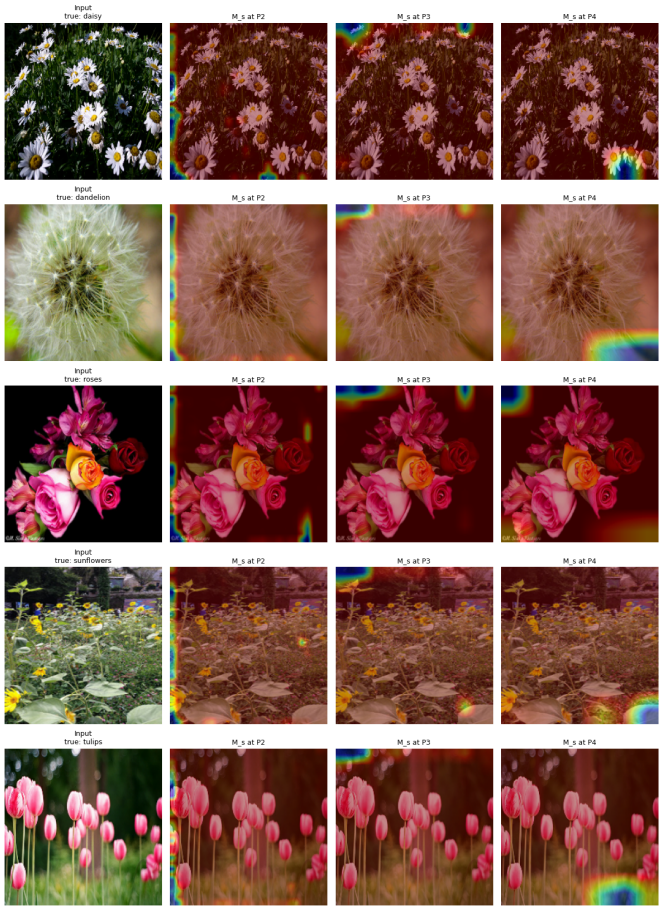


Inference: The response collapses onto one or two small bright spots per image and the center query attends to nearly the same spot as the mean and maximum maps, so long-range interaction happens mainly through only a few locations.

14.4 Attention at Different ResNet Stages

For the same test images, $M_s^{P_2}$, $M_s^{P_3}$ and $M_s^{P_4}$ are shown.

Spatial Attention at P_2, P_3 and P_4



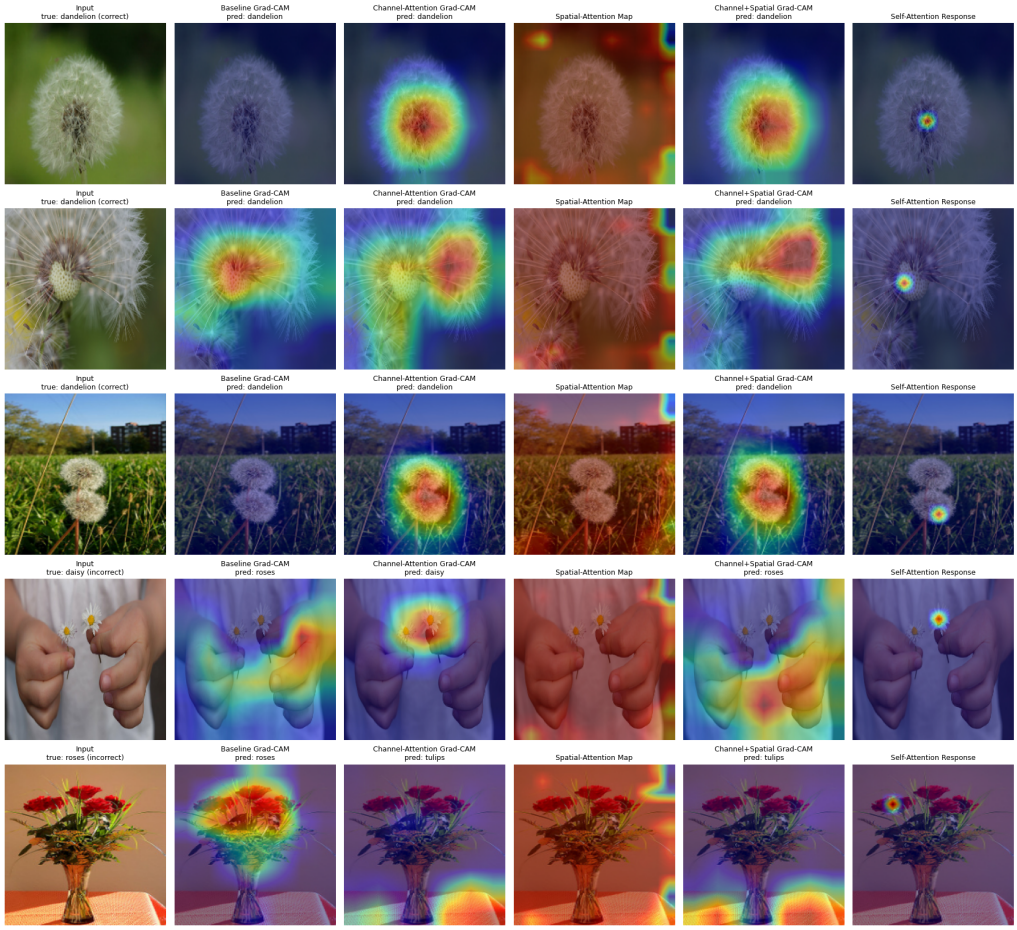
Inference: The maps become coarser from P_2 (28×28) to P_4 (7×7) but all stay close to 1 almost everywhere (low only at borders or one corner block), so they do not show the expected edge to part to semantic progression.

15. Qualitative Comparison

Three correctly classified and two incorrectly classified test images are compared across models (same images for all models).

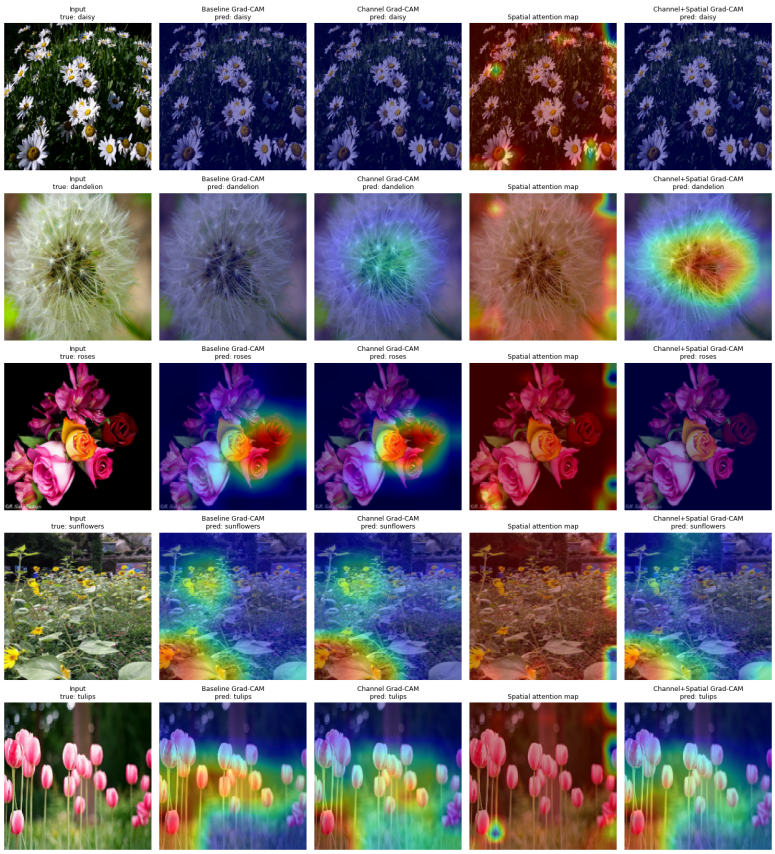
Input | Baseline Grad-CAM | Channel Grad-CAM
Spatial Attention | Channel+Spatial Grad-CAM | Self-Attention

Qualitative Comparison Across Models



Inference: For the correct dandelion images the channel and Channel+Spatial models are compact on the flower head while the baseline is weak or diffuse. For the daisy held in hands the baseline and A3 predict roses (activation on hands/torso) but the channel model correctly predicts daisy on the flower and for the rose bouquet the attention models predict tulips with activation on the table and vase, so wrong evidence accompanies wrong predictions.

Recommended Layout (one image per class)



Inference: The channel and Channel+Spatial Grad-CAMs are similar (compact on the dandelion, comparable to the baseline on tulips and sunflowers) while the spatial map is nearly flat, so localization comes from the channel/classifier path rather than the spatial gate .

Criteria: foreground localization is good on dandelions and weak on cluttered images. Maps focus on the flower centre rather than individual petals, attention Grad-CAMs are more compact but the spatial gate covers almost the whole image. Channel/spatial gates are consistent across images whereas Grad-CAM and self-attention change per image.

16. Source Code

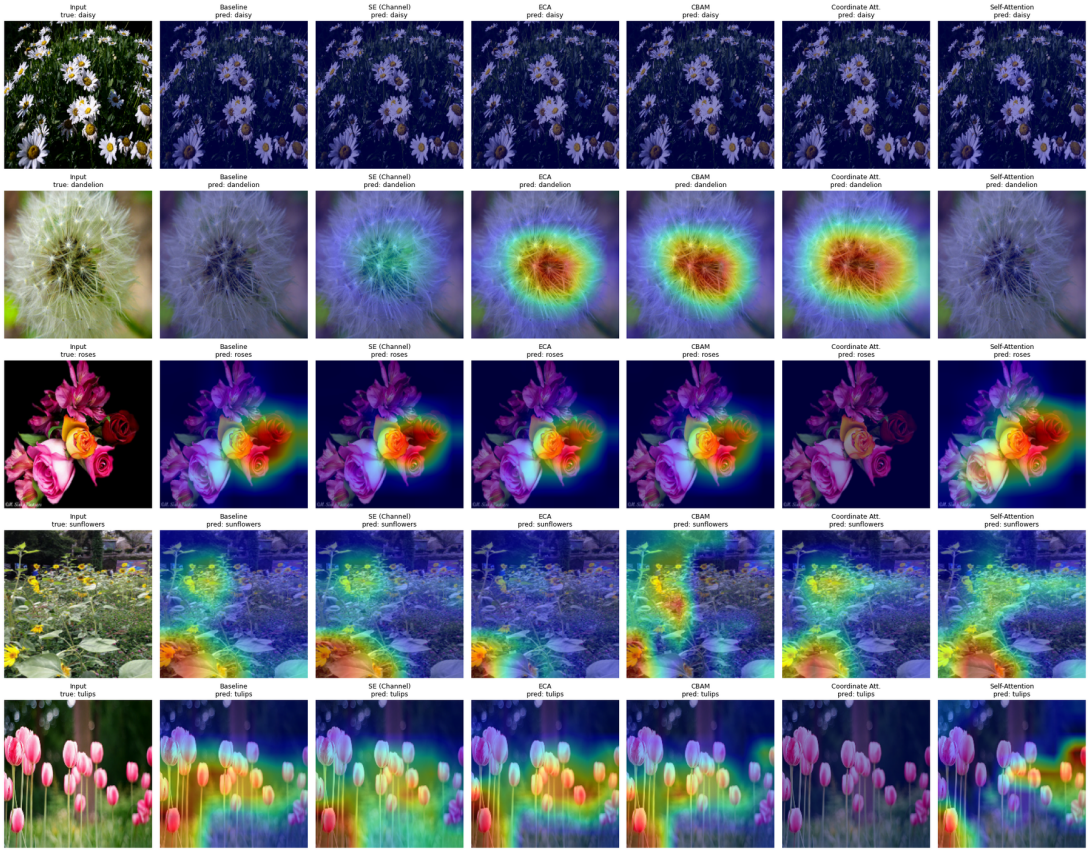
GitHub Repository

17. Additional Attention-Mechanism Comparison

ECA, CBAM and Coordinate attention were compared with SE (same as channel attention A1) and self-attention, keeping ResNet50 and the insertion position P_3 fixed.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
No Attention	–	0.9347	0.9355	0.9341	0.9345	23,597,957	7.25 ms
Channel / SE	P_3	0.9220	0.9292	0.9189	0.9213	23,730,117	8.01 ms
Spatial Attention	P_3	0.9419	0.9393	0.9449	0.9411	23,598,056	7.44 ms
ECA	P_3	0.8857	0.8996	0.8861	0.8841	23,597,962	7.32 ms
CBAM	P_3	0.9002	0.9107	0.8949	0.8970	23,730,216	7.93 ms
Self-Attention	P_3	0.9238	0.9230	0.9241	0.9219	24,909,958	6.74 ms
Coordinate Attention	P_3	0.9274	0.9296	0.9270	0.9282	23,698,469	7.65 ms

Extended Comparison (Grad-CAM for Baseline, SE, ECA, CBAM, Coordinate, Self-Attention)



Inference: On the dandelion ECA, CBAM and Coordinate attention concentrate the activation on the flower head while SE shows only a weak ring and Self-Attention is as diffuse as the baseline .On tulips and sunflowers all maps stay broad. ECA and CBAM gave the lowest accuracy so compact Grad-CAM did not translate into better classification.

Comparisons

1. **Accuracy:** from 0.8857 (ECA) to 0.9419 (Spatial); baseline 0.9347.
2. **Macro F1:** 0.8841 (ECA) to 0.9411 (Spatial), same ranking as accuracy.
3. **Added parameters:** SE 132,160; Spatial 99; ECA 5; CBAM 132,259; Self-Attention 1,312,001; Coordinate 100,512.
4. **Inference time:** 6.74–8.01 ms/image versus 7.25 ms for the baseline, with no relation to module size (timing noise).
5. **Convergence:** Self-Attention and Coordinate attention fit the training data faster but validate the same; ECA converges slowest.

6. **Focused localization:** only on some images (dandelion), not on tulips or sunflowers.
7. **Does more complexity always help?** No: Self-Attention (+1.31M) reached 0.9238, below the baseline.
8. **Lightweight vs heavy:** Spatial (99 parameters) was best, but ECA (5 parameters) was worst, so lightness alone is not enough.

18. Discussion

Key discussion points

- **Why performance changed:** The ImageNet backbone was already strong (93.5%) so training extra parameters on only 2,569 images yielded little benefit. Only the spatial gate helped, while ECA/CBAM proved unstable.
- **Location and resolution:** Early insertion at P_1/P_2 reduced accuracy by disrupting low-level features that later stages expect, whereas P_3/P_4 remained neutral with mostly flat spatial maps.
- **Complexity:** Adding up to 1.44M parameters yielded no matching performance gains, meaning only Spatial attention (+0.7 pp) is practical.
- **Maps and background:** Internal maps stay largely diffuse. Grad-CAM focuses on flower heads in some cases (dandelion) but often wanders to stems, background clutter, or hands in misclassifications.
- **Quantitative vs qualitative:** Visual appeal did not track accuracy (tighter ECA/CBAM maps failed to increase scores).

Answers to the questions mentioned

1. **What is the purpose of residual connections in ResNet50?**

   They provide an identity shortcut ($y = F(x) + x$) that lets gradients backpropagate directly, preventing degradation in deep networks.

2. **What does conv2_x represent compared with conv5_x?**

   `conv2_x` captures high-resolution local edges and textures (56×56), while `conv5_x` extracts coarse class-specific semantic patterns (7×7).

3. **Why does the spatial resolution decrease with network depth?**

   Strided operations expand the effective receptive field and lower computation trading spatial precision for broader context.

4. **Why does the number of feature channels increase with network depth?**

   Deeper layers need more channels to encode a richer set of complex parts and category concepts, offsetting the lost spatial dimensions.

5. **What is meant by low-level, intermediate, and high-level features?**

   Low-level captures edges and colors while intermediate captures recurring textures and petals. High-level encodes the entire flower structure.

6. **What is the purpose of an attention mechanism?**

   To dynamically emphasize informative features while suppressing background noise, though it does not guarantee higher classification accuracy.

7. **What does channel attention attempt to learn?**

   It learns a 1D importance weight per feature map to decide what specific channels are most informative.

8. **What does spatial attention attempt to learn?**

   It learns a 2D spatial weight map across the feature grid to prioritize where key information is located.

9. **How is channel attention different from spatial attention?**

   Channel attention produces a C-dimensional vector to re-weight feature maps, whereas spatial attention outputs an $H \times W$ map across locations.

10. **Why can channel and spatial attention be used sequentially?**

    They target orthogonal axes allowing the model to refine feature types first and then spatial regions (or vice versa).

11. **Why might channel attention behave differently at conv3_x and conv5_x?**

    `conv3_x` encodes generic mid-level textures while `conv5_x` encodes semantic classes. Here, P_3 weights simply saturated near 1.

12. **Why can spatial attention be useful at an intermediate ResNet stage?**

    Resolutions like 28×28 preserve enough spatial detail to outline foreground objects while carrying useful part-level semantics.

13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**

    It calculates pairwise interactions across all $N = HW$ positions, causing both memory footprint and computation to scale as $\mathcal{O}(N^2)$.

14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**

    It expands quadratically (N^2), growing from 2,401 entries at P_4 to nearly 9.8×10^6 at P_1, a 16 times jump every time dimensions double.

15. **How does attention performance vary when the insertion position is changed?**

    Inserting modules early at P_1/P_2 lowered accuracy to ∼90.4%, whereas late insertion at P_3/P_4 proved much safer (∼93.5%).

16. **How does attention at P_2 differ visually from attention at P_4?**

    P_2 maps are higher-resolution (28×28) with border attenuation, while P_4 maps are coarse 7×7 blocks neither cleanly traces the flower.

17. **Does applying attention at multiple stages always improve classification performance?**

    No cascading $P_3 + P_4$ added minimal gain (+0.36 pp) at 5 times the parameters while $P_2 + P_3 + P_4$ dropped accuracy to 0.9292.

18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**

    Hardly. They scored 0.9310 and 0.9328 respectively a negligible change.

19. **How does the spatial-attention kernel size influence the attention map?**

    Larger kernels smooth out the map with more context but testing $k \in \{3, 5, 7\}$ yield similar accuracies with no measurable upside .

20. **How does SE attention differ from ECA attention?**

    SE uses global pooling and a two-layer bottleneck MLP (132k params), whereas ECA uses an efficient 1D convolution (5 params).

21. **How does CBAM combine channel and spatial attention?**

    It routes pooled descriptors through a shared MLP for channel weighting then applies a 7×7 spatial convolution (reaching 0.9002 here).

22. **How does self-attention differ from channel attention?**

    Self-attention calculates pairwise spatial affinities via an $N \times N$ matrix, while channel attention outputs a single scalar per channel .

23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**

    No . Self-Attention added 1.31M parameters but dropped accuracy (0.9238), while a tiny 99-parameter spatial gate reached 0.9419.

24. **What is the difference between an internal attention map and Grad-CAM?**

    An internal map is a forward-pass tensor that directly modulates features whereas Grad-CAM is a post-hoc gradient-based explanation tool.

25. **Why is Grad-CAM called a post-hoc explanation technique?**

    It probes a trained model's predictions after inference via backward gradients without altering the underlying architecture or training.

26. **Does the attention model suppress irrelevant background regions?**

    Only partially as it tightens focus around dandelions but other classes still trigger on leaves, soil or image borders.

27. **Are the attention maps of correctly classified and incorrectly classified images different?**

    Yes. Correct predictions center more on flower heads while errors frequently latch onto hands, vases or tabletops.

28. **Which flower classes benefit most from attention?**

    Sunflowers saw the highest F1 improvement (+0.008) and roses gained recall (+4.2 pp) whereas tulips and dandelions declined slightly.

29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**

    It uncovers pairwise error directions and class-specific confusions (like roses mistaken for tulips) that a summary metric cannot determine.

30. **Why is macro F1 useful in addition to accuracy?**

    It evaluates all classes with equal weight ensuring weak performance on underrepresented or difficult classes is not masked.

31. **What is meant by the accuracy–complexity trade-off?**

    It weighs whether accuracy gains justify extra compute and memory. Here, additional parameter overhead yielded no real change.

32. **What is meant by the F1–inference-time trade-off?**

    It measures whether gains in balanced performance justify slower runtime. Here, higher latencies brought no F1 improvements.

33. **Why must every ablation study use the same dataset split?**

    It ensures all observed score differences come strictly from architectural changes rather than shifting test set difficulty.

34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**

    It isolates the attention module as the single independent variable preventing optimizers or augmentations from skewing results.

35. **What conclusions can be drawn from the Grad-CAM comparison?**

    Attention can shift visual focus and errors align with background activation but these patterns don't track accuracy and are only qualitative.

19. Conclusion

This experiment shows that attention effectiveness depends on the feature representation. Based on performance, computational complexity, and visualization, spatial attention after P_3 performed best, achieving 94.19% accuracy and 0.9411 macro F1 with only 99 extra parameters. Other attention configurations offered little or no improvement. Visualizations revealed image-dependent attention, occasional background activation, and off flower evidence in misclassifications. However all performance differences fall within the approximately ± 2.1.

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